# POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Basic Electrical Engineering (New)

Semester: Spring

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

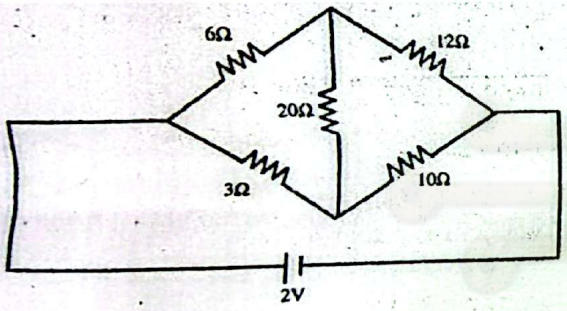
*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

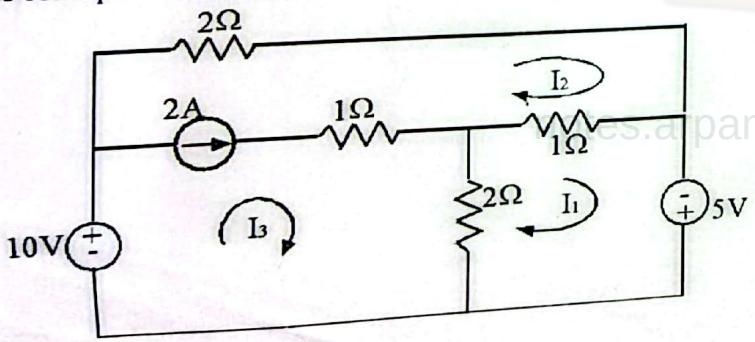
***Attempt all the questions.***

1. a) What are the sources of electricity? Explain generation of electricity from Hydro. 7

   b) Determine the current in 20 Ω resistor of the network shown in figure below using Star-Delta Transformation. 8

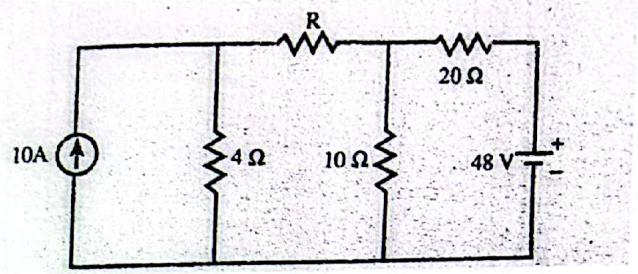



2. a) Find the mesh current $I_1$, $I_2$ and $I_3$ in the circuit given below using the concept of mesh analysis. 8



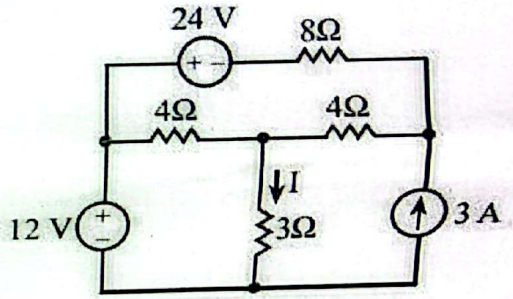



b) What is the value of R such that maximum power is transferred to it? Find the value of this maximum power for the circuit shown below. 7

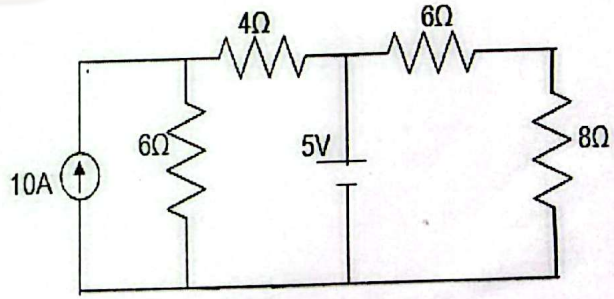



**OR**

Use Superposition theorem to find current flowing through the 3 ohms resistor shown in the figure.



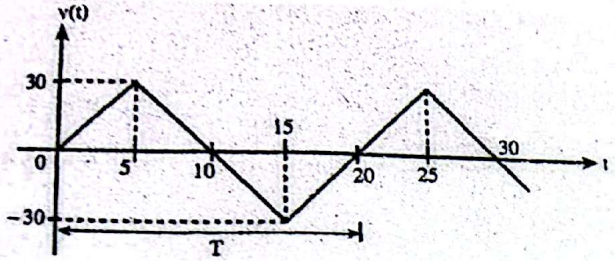

3. a) Calculate the power dissipation across 4Ω resistor using Norton's theorem. 8






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b) Determine the average and rms value of voltage for waveform as shown in figure below. 7




4. a) Two impedances (200+j25) Ω and (100-j8) Ω are connected in parallel across a 430V, 50Hz supply. Find active power, reactive power, apparent power and power factor of the circuit. 7

b) Three arms of a three phase load each comprise resistance of 25Ω and inductance of 0.15H in series. The three phase supply is 415V, 50Hz. Calculate line current, power factor and the total power in KW when the load is 8

i. star connected

ii. mesh connected.

5. a) Explain how we can generate three phase voltage. Why three phase system is preferred over single phase system. 7

b) *A 10-kVA, 250V/500V, 50-Hz, single-phase transformer gave the following test results:* 8

*OC test (HV winding open) : 250 V, 1 A, 80 W.*
*SC test (LV winding shorted) : 20 V, 12 A, 100 W.*
*Find the equivalent circuit parameters as referred to*

i. *LV side*

ii. *HV side*

6. a) Explain the construction and operation of three phase induction motor along with their uses. 7

b) A 230V dc shunt motor has an armature resistance of 0.1 ohm and a shunt field resistance of 275 ohm. It runs at speed of 1000 rpm when drawing an armature current of 75A. Calculate the additional resistance to be inserted in the field circuit to raise the motor speed to 1200 rpm at an armature current of 125A. 8

**OR**



A 200 V d.c. series motor runs at 1,000 rpm, and takes 20 A. Combined resistance of field and armature is 0.4 Ω. Calculate the resistance to be inserted in series so as to reduce the speed to 800 rpm, assuming torque varies as the square of speed.

7. Write short notes on: (**Any two**) 2×5

a) Back Emf

b) Power factor and its significance

c) Kirchhoff's law



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